

CYBERSECURITY INTERNSHIP

FINAL TECHNICAL SUBMISSION

1.1 Networking Fundamentals • 1.2 Linux & System Security • 1.3 Threat & Vulnerability Identification •
1.4 Basic Security Assessment

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Program	B.Tech — Cyber Security
Level	Beginner → Intermediate
Lab Environment	Kali Linux / Controlled Virtual-Machine Lab
Submission	Internship Deliverables 1.1–1.4
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Submission note: Topic 1.2 includes the supplied Linux notes and terminal evidence as an appendix. Topic 1.4 is documented as a controlled-lab assessment procedure unless execution evidence is available; no unverified scan results are claimed.

1.1 Networking Fundamentals — BEGINNER

Objective

Understand the TCP/IP model and the roles of TCP, UDP, DNS and HTTP, then connect these concepts to common ports encountered during security analysis.

TCP/IP Model

Application	HTTP/HTTPS • DNS • SSH • SMTP • DNS queries
Transport	TCP • UDP
Internet	IP • ICMP
Network Access	Ethernet • Wi-Fi • ARP

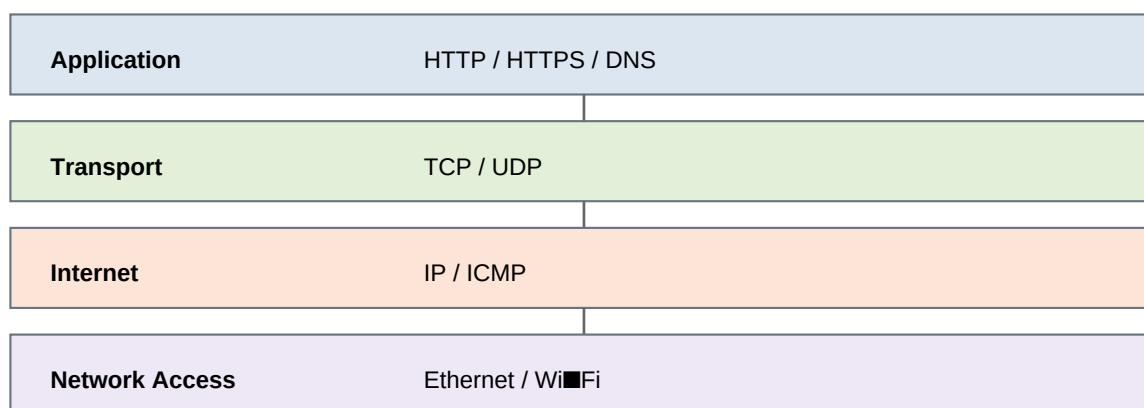


Diagram 1. TCP/IP stack showing where the key security-analysis protocols fit.

TCP

TCP is a transport-layer protocol that provides reliable, ordered communication between endpoints. It uses connection establishment and sequence/acknowledgement mechanisms. TCP is commonly encountered with services such as HTTP/HTTPS and SSH. The current TCP specification is RFC 9293.

UDP

UDP is a lightweight transport protocol that provides datagram-based communication with minimal protocol mechanism. It does not provide TCP-style reliability or ordering. DNS commonly uses UDP for ordinary queries, while some applications use UDP for latency-sensitive communication. RFC 768 defines UDP.

DNS

The Domain Name System maps domain-style names to resource information and uses resolvers and name servers. DNS records can include address and mail-related information. DNS concepts are described in RFC 1034 and implementation/protocol details in RFC 1035.

HTTP / HTTPS

HTTP is a stateless application-level request/response protocol. A client sends a request to a server and the server returns a response with status and content. HTTPS uses the https URI scheme and a secured connection. HTTP semantics are defined by RFC 9110.

Common Ports Relevant to Security Analysis

Port	Typical service	Security-analysis relevance
20/21 TCP	FTP / data-control	Legacy file transfer; review exposure and authentication.
22 TCP	SSH	Remote administration; check authentication and configuration.
25 TCP	SMTP	Mail transfer; identify unnecessary exposure.
53 TCP/UDP	DNS	Name resolution; inspect DNS behavior and exposure.
80 TCP	HTTP	Web service; inspect HTTP headers, configuration and application behavior.
110 TCP	POP3	Legacy mail retrieval; secure alternatives should be preferred where appropriate.
143 TCP	IMAP	Mail retrieval; review encryption/authentication configuration.
443 TCP	HTTPS	Encrypted web service; inspect TLS and application security.
3306 TCP	MySQL	Database service; should not normally be unnecessarily internet-exposed.
3389 TCP	RDP	Remote desktop; review exposure and authentication controls.

Security workflow: identify the host → identify reachable services/ports → identify the protocol → review configuration and authentication → correlate findings with vulnerabilities and business impact.

LINUX COMMANDS

Beginner-Friendly Internship & Cybersecurity Notes

Focus: Command line basics, files & directories, permissions, users, processes, searching, networking and basic security.

Prepared for Cybersecurity Learning

1. Objective

- ❑ The objective of this lab is to develop practical knowledge of Linux command-line operations related to user management, group management, permissions, and basic system security.
- ❑ The exercises help in understanding how Linux identifies users, manages accounts and groups, and controls access to system resources

Command	
whoami	Displays the current logged-in username.
id	Displays UID, GID and group memberships.
hostname	Displays the system hostname.
uname -a	Displays detailed kernel/system information.
pwd	Displays the current working directory
cat	Displays file contents
grep	Search/filter lines matching a text or pattern
(pipe)	Passes one command's output as input to another command
useradd -m	Creates a user and its home directory
passwd	Sets or changes a user's password
groups	Displays group membership
groupadd	Creates a new group

1. Linux Command Line Basics

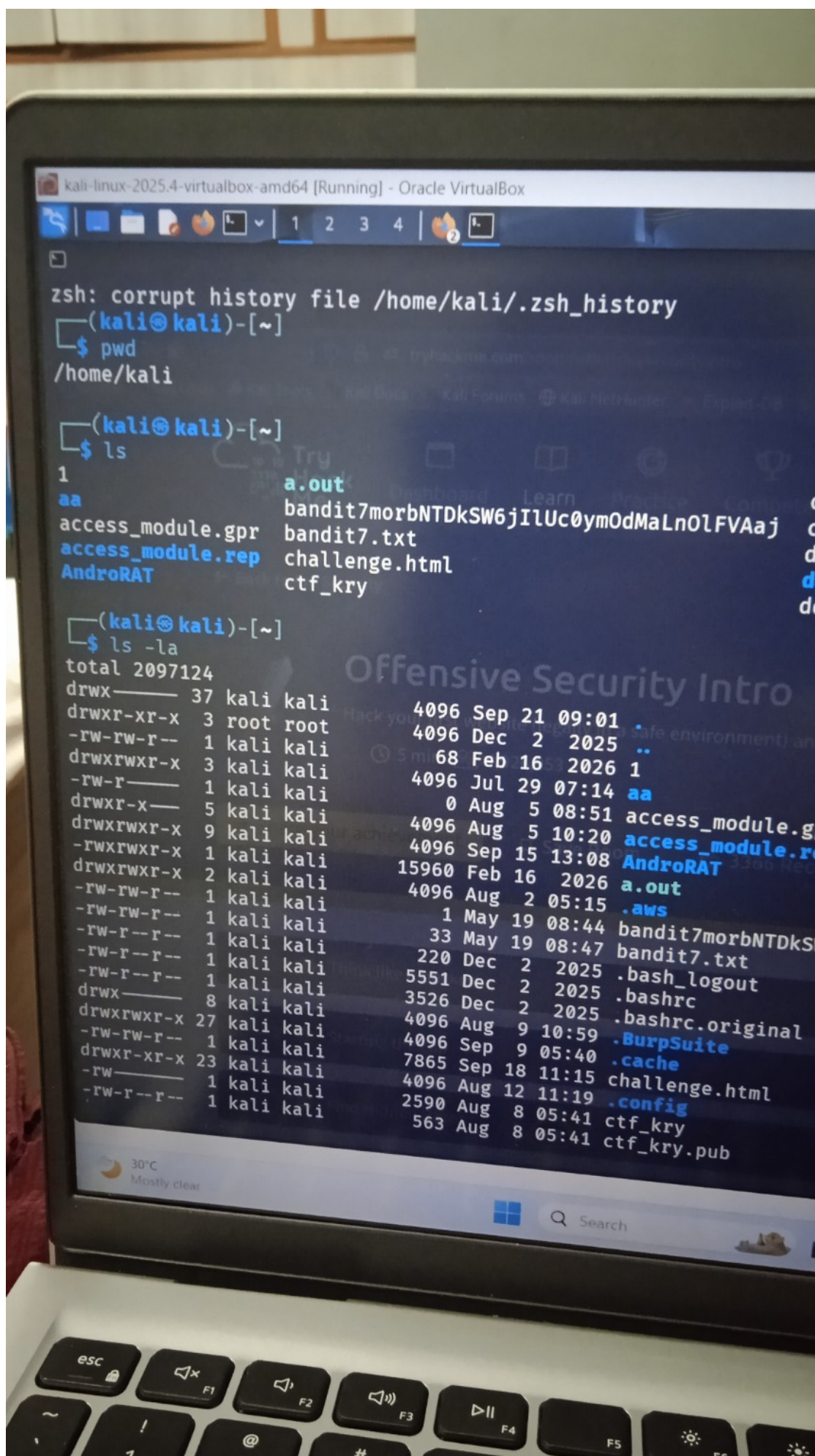
Command	Purpose	Example
pwd	Print Working Directory — shows the directory you are currently in.	pwd
ls	Lists files and directories.	ls ls -l ls -la
cd	Changes the current directory.	cd Documents cd .. cd ~
clear	Clears the terminal screen.	clear
whoami	Displays the currently logged-in username.	whoami
date	Displays the current date and time.	date
history	Shows previously executed commands.	history

2. Files and Directories

Command	Purpose	Example
mkdir	Creates a new directory.	mkdir practice
touch	Creates an empty file or updates its timestamp.	touch notes.txt
cp	Copies files or directories.	cp notes.txt backup.txt cp -r folder/ backup/
mv	Moves or renames files/directories.	mv old.txt new.txt
rm	Removes a file. Use carefully.	rm notes.txt rm -r folder/
rmdir	Removes an empty directory.	rmdir empty_folder

3. Viewing File Content

Command	Purpose	Example
cat	Displays file contents.	cat notes.txt
less	Views long files page by page.	less /var/log/syslog
head	Shows the beginning of a file.	head notes.txt head -n 10 notes.txt
tail	Shows the end of a file.	tail notes.txt tail -n 20 notes.txt
wc	Counts lines, words and bytes.	wc notes.txt wc -l notes.txt



4. Permissions and Ownership

Command	Purpose	Example
ls -l	Displays permissions, owner and group.	<code>ls -l</code>
chmod	Changes file permissions.	<code>chmod 644 notes.txt</code> <code>chmod +x script.sh</code>
chown	Changes file owner/group (requires appropriate privileges).	<code>sudo chown user:user notes.txt</code>
chgrp	Changes the group ownership.	<code>sudo chgrp students notes.txt</code>

5. Users and Groups

Command	Purpose	Example
id	Displays user and group IDs.	<code>id</code>
who	Shows users currently logged in.	<code>who</code>
w	Shows logged-in users and activity.	<code>w</code>
groups	Shows groups associated with a user.	<code>groups</code>
passwd	Changes a user's password (subject to system policy).	<code>passwd</code>

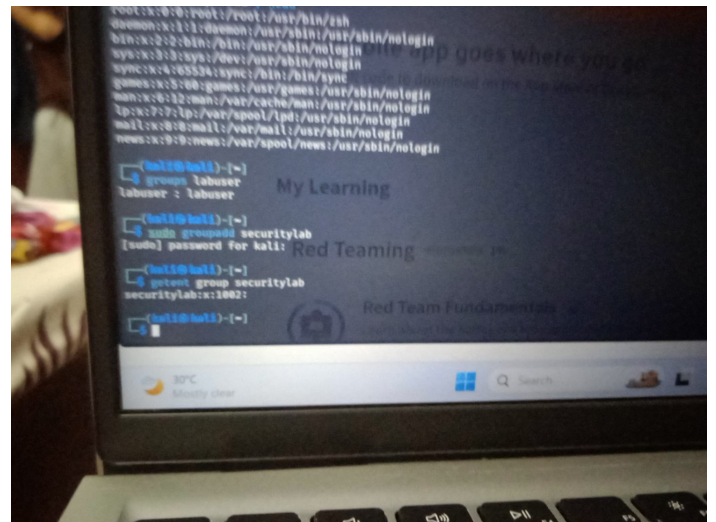
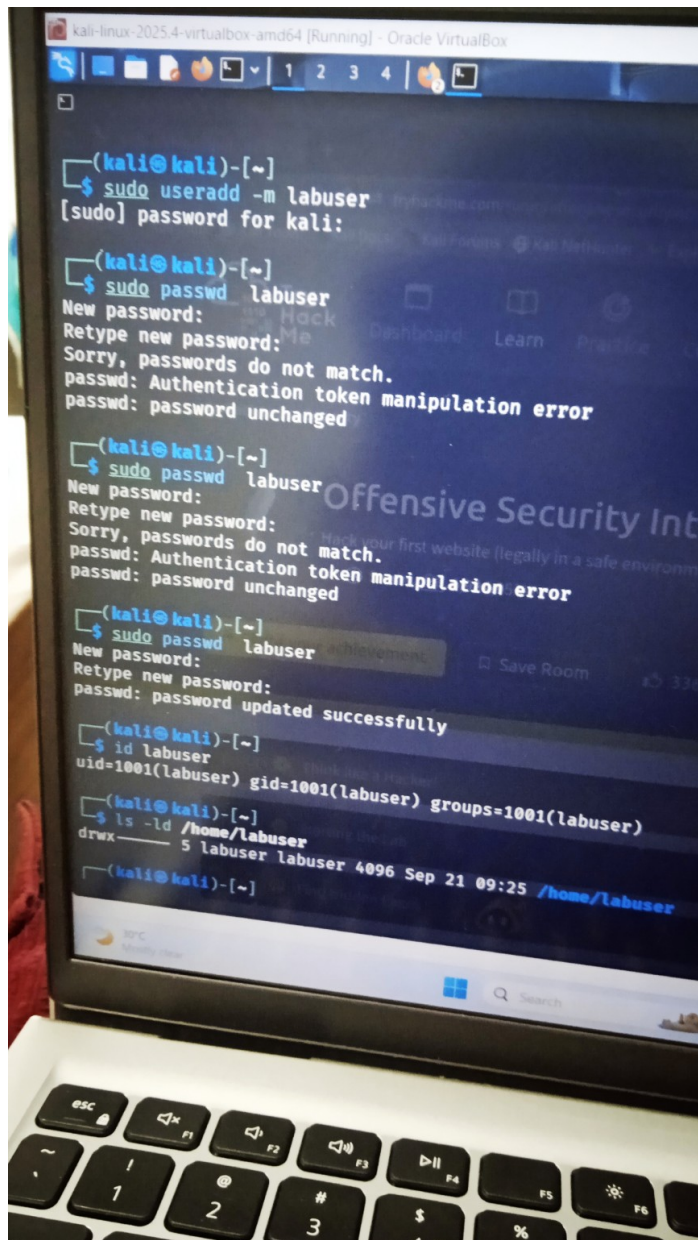
6. Searching and Filtering

Command	Purpose	Example
grep	Searches for a text pattern inside input/files.	<code>grep "error" app.log</code> <code>grep -i "failed" auth.log</code>
find	Searches for files/directories by conditions.	<code>find . -name "*.txt"</code> <code>find /tmp -type f</code>
locate	Quickly searches a filename database when available.	<code>locate notes.txt</code>
sort	Sorts lines of text.	<code>sort names.txt</code>
uniq	Removes adjacent duplicate lines; often used with sort.	<code>sort names.txt uniq</code>

7. Pipes and Redirection

Command	Purpose	Example
 pipe	Sends the output of one command as input to another.	<code>ls -la grep "txt"</code>

Linux Practical Evidence — Users & Groups



Command	Purpose	Example
>	Writes command output to a file, replacing existing content.	<code>ls > files.txt</code>
>>	Appends command output to a file.	<code>date >> log.txt</code>
<	Uses a file as standard input.	<code>sort < names.txt</code>
tee	Displays output and also writes it to a file.	<code>ls -la tee listing.txt</code>

8. Processes and System Information

Command	Purpose	Example
ps	Displays running processes.	<code>ps</code> <code>ps aux</code>
top	Interactive view of running processes and resource usage.	<code>top</code>
kill	Sends a signal to a process by PID.	<code>kill PID</code>
free	Displays memory usage.	<code>free -h</code>
df	Shows filesystem disk space.	<code>df -h</code>
du	Shows file/directory disk usage.	<code>du -sh folder/</code>
uname	Displays system/kernel information.	<code>uname -a</code>

9. Networking Commands

Command	Purpose	Example
ip	Displays and manages network interfaces/routes.	<code>ip addr</code> <code>ip route</code>
ping	Tests network reachability using ICMP where permitted.	<code>ping -c 4 8.8.8.8</code>
ss	Displays sockets and network connections/listeners.	<code>ss -tuln</code>
curl	Transfers data and can make HTTP requests.	<code>curl https://example.com</code>
wget	Downloads content from supported URLs.	<code>wget https://example.com/file.txt</code>
nslookup	Queries DNS information.	<code>nslookup example.com</code>
dig	Detailed DNS query tool, if installed.	<code>dig example.com</code>

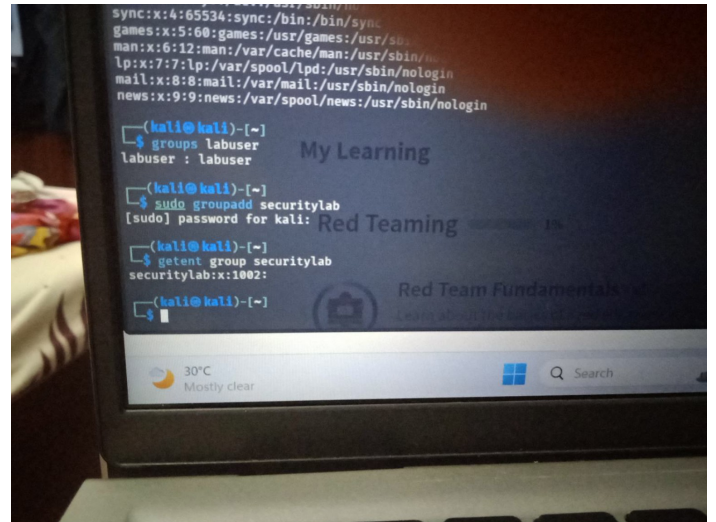
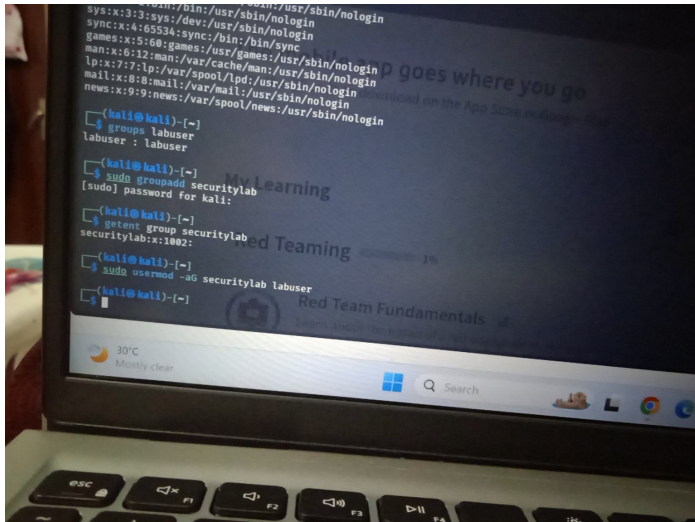
10. Archives and Compression

Command	Purpose	Example
tar	Creates or extracts tar archives.	tar -cvf archive.tar folder/ tar -xvf archive.tar
gzip	Compresses a file.	gzip notes.txt
gunzip	Decompresses a .gz file.	gunzip notes.txt.gz
zip/unzip	Creates/extracts ZIP archives when installed.	zip -r files.zip folder/ unzip files.zip

11. Useful Cybersecurity Commands

Command	Purpose	Example
sha256sum	Calculates a SHA-256 hash for integrity checking.	sha256sum file.iso
file	Identifies a file type from its contents.	file suspicious.bin
strings	Extracts printable strings from binary-like files.	strings sample.bin
hexdump	Displays a file in hexadecimal/ASCII form.	hexdump -C sample.bin
journalctl	Queries systemd journal logs on systems using systemd.	journalctl journalctl -u ssh
sudo	Runs a command with elevated privileges when authorized.	sudo command

Linux Practical Evidence — /etc/passwd & Group Commands



TOPIC – 2

LINUX & SYSTEM SECURITY EXERCISES

Linux Command-Line Exercises

Permissions, Users & Basic Hardening

1. Linux User Identification

Linux provides several commands to identify the current user and understand the system environment.

- **Command:** `whoami`
- **Purpose:** The `whoami` command displays the `username` of the currently logged-in user.

This means the current terminal session is running under the `kali` user.

❑ **Command:** id

➤ **Purpose:** The id command displays:

User ID (UID)

Group ID (GID)

Group memberships

❑ **Example:**

uid=1000(kali) gid=1000(kali) groups=1000(kali),27(sudo)

➤ **Explanation:**

UID → Unique numerical ID assigned to a user.

GID → ID of the user's primary group.

Groups → Groups to which the user belongs.

Security relevance:

A security analyst can use id to determine the privilege and group context of a user.

2. System Information

➤ **Command:**

hostname

Purpose:

Displays the name of the Linux machine.

➤ **Command:**

uname -a

Purpose:

Displays detailed information about the Linux kernel and system architecture.

➤ **Command:**

pwd

Purpose:

pwd means Print Working Directory.

It displays the directory in which the user is currently working.

Example:

/home/kali

3. Understanding /etc/passwd

Linux stores basic local user-account information in:
/etc/passwd

To display it:

cat /etc/passwd

The cat command displays the contents of a file.

A typical entry looks like:

```
kali:x:1000:1000:Kali User:/home/kali:/bin/bash
```

The fields represent:

Username : Password placeholder : UID : GID : Comment : Home Directory : Login Shell

The actual password information is not stored directly in /etc/passwd; protected password data is normally stored in /etc/shadow.

5. grep Command

What is grep?

grep is used to search for specific text or patterns inside files or command output.

Example:

```
grep "kali" /etc/passwd
```

This searches /etc/passwd for lines containing kali.

Useful concept: **cat file.txt | grep "word"**

Here:

cat → displays the file

| → pipe; sends output to the next command

grep → searches/filter the output

Example used in the lab:

```
grep "^$(whoami):" /etc/passwd
```

Here:

^ → beginning of the line

\$(whoami) → runs whoami and uses its output

: → separates fields in /etc/passwd

6. Creating a Linux User

A separate test account was created for the security lab.

Command:

```
sudo useradd -m labuser
```

Explanation:

sudo → executes the command with administrative privileges.

useradd → creates a new user.

-m → creates the user's home directory.

labuser → name of the new account.

The user's home directory is normally:

/home/labuser

Security relevance:

Using a separate test account is safer for practical exercises because it avoids unnecessarily modifying the main account.

7. Setting the User Password

Command:

sudo passwd labuser

Purpose: The passwd command is used to set or change a user's password.

The terminal asks for the password twice.

Important: Linux normally does not display password characters while typing. This is expected behavior.

8. Checking User Information

Command: **id labuser**

Purpose: Displays the UID, GID and group membership of the specific user labuser.

Example: uid=1001(labuser) gid=1001(labuser) groups=1001(labuser) This shows that the user has its own UID and primary group.

9. Linux Groups

A group is a collection of users.

Groups are useful for managing permissions for multiple users.

Command: `groups labuser`

Purpose: Displays the groups to which labuser belongs.

Initially, the output was:

labuser : labuser

This means the user belonged to its primary group.

10. Creating a New Group

A security-focused group was created for the lab.

Command: `sudo groupadd securitylab`

Explanation:

`sudo` → administrative privileges

`groupadd` → creates a new group

`securitylab` → name of the group

The group was verified using:

`getent group securitylab`

Example output:

`securitylab:x:1002:`

This confirmed that the group was successfully created.

11. Adding a User to a Group

The labuser account was added to the securitylab group.

Command: `sudo usermod -aG securitylab labuser`

Explanation: `usermod` Used to modify an existing user account.

-G Specifies supplementary groups.

-a Means append.

It adds the new group while preserving existing supplementary group memberships.

Securitylab

The group to which the user is being added.

labuser

The user being modified.

Security importance:

Group membership can affect what resources a user can access. Therefore, unnecessary or excessive group memberships should be reviewed during security audits.

12. File Permissions — Basic Concept

Linux uses permissions to control access to files and directories.

There are three main categories:

Owner

Group

Others

And three basic permissions:

r = Read

w = Write

x = Execute

Numeric values:

Permission	Value
s	
Read(r)	4
Write(w)	2
Execute(x)	1

Means: Owner → rwx

Group → r-x

Others → r-x

Permissions are important because incorrect permissions can expose sensitive information or allow unauthorized modification.

13. Basic Hardening NotesBasic

Linux hardening includes:

Use strong passwords.

Keep the operating system updated.

Remove unnecessary user accounts.

Review group memberships.Disable unnecessary services.Use least-privilege access.Protect sensitive files with appropriate

permissions.Monitor login and authentication activity.Review listening network services.Use a firewall where appropriate.

1.3 Threat & Vulnerability Identification — BEGINNER

Purpose

Identify common threats, vulnerabilities and attack vectors relevant to web systems, and understand the basic defensive control for each.

Threat vs Vulnerability vs Attack Vector

Term	Meaning	Example
Threat	A potential cause of harm to a system or data.	An attacker attempting to steal account data.
Vulnerability	A weakness that can be exploited.	Missing authorization check on an admin endpoint.
Attack vector	The path or mechanism used to exploit a weakness.	Crafted HTTP request, malicious input, or stolen credential.
Risk	Potential impact considering likelihood and consequence.	Unauthorized data access causing confidentiality loss.

Broken Access Control

What it is: Users can access or modify resources beyond their authorization.

Basic defense: Enforce server-side authorization on every sensitive operation; deny by default; test object-level access.

Security Misconfiguration

What it is: Unsafe defaults, exposed admin interfaces, excessive permissions, or unnecessary services.

Basic defense: Harden configurations, remove defaults, minimize services, review permissions and security headers.

Injection

What it is: Untrusted input is interpreted as commands or queries.

Basic defense: Use parameterized queries, safe APIs, strict input validation and output encoding.

Authentication Failures

What it is: Weak authentication or session handling allows account compromise or session abuse.

Basic defense: Use strong authentication, secure session management, MFA where appropriate, rate limits and secure credential handling.

Cryptographic Failures

What it is: Sensitive data is inadequately protected in storage or transit.

Basic defense: Use appropriate modern cryptography, TLS, key management and avoid unnecessary sensitive-data retention.

1.3 Threat & Vulnerability Identification — continued

Additional common web risks

Software Supply Chain Failures

What it is: Compromise or weakness in dependencies, build systems or distribution can affect applications.

Basic defense: Inventory dependencies, verify trusted versions, monitor advisories and secure build pipelines.

Software/Data Integrity Failures

What it is: The system fails to verify that code or data came from a trusted source or was not altered.

Basic defense: Use integrity verification, signed artifacts and trusted update mechanisms.

Logging & Alerting Failures

What it is: Important security events are not recorded or alerted on effectively.

Basic defense: Log relevant security events, protect logs, monitor alerts and define response procedures.

Mishandling Exceptional Conditions

What it is: Unexpected conditions lead to unsafe behavior, logical errors or fail-open states.

Basic defense: Use safe error handling, validation, fail-closed behavior where appropriate and security-focused testing.

OWASP Top 10:2025 — reference list

ID	Category
A01	Broken Access Control
A02	Security Misconfiguration
A03	Software Supply Chain Failures
A04	Cryptographic Failures
A05	Injection
A06	Insecure Design
A07	Authentication Failures
A08	Software or Data Integrity Failures
A09	Security Logging & Alerting Failures
A10	Mishandling of Exceptional Conditions

Safe learning rule: perform testing only on systems you own or are explicitly authorized to assess. For internship practice, use a local VM or intentionally vulnerable training target.

1.4 Basic Security Assessment — INTERMEDIATE

Controlled / Local Virtual-Machine Assessment

A basic assessment should be performed against a controlled target such as a local virtual machine or an intentionally vulnerable lab. The aim is to identify exposed services, basic configuration weaknesses and obvious web/application issues without disrupting the target.

Assessment Scope

Phase	Goal	Example evidence
1. Scope	Define the target and authorization boundary.	Target IP/hostname and lab description.
2. Host discovery	Confirm the target is reachable.	Ping or equivalent local connectivity check.
3. Port/service discovery	Identify listening services and versions.	Nmap output; ss output on the VM.
4. Web enumeration	Identify web service behavior.	HTTP headers, status code, technology clues.
5. Configuration review	Check users, permissions, services and firewall.	Linux command output.
6. Vulnerability review	Map observations to known weaknesses.	OWASP/NVD/vendor references.
7. Evidence	Record commands, timestamps and observations.	Screenshots and terminal logs.
8. Remediation	Recommend specific fixes and verification.	Finding → risk → fix → retest.

Safe lab command checklist

Use the following only against your controlled/local VM target:

```
ip addr
ip route
ping -c 4 127.0.0.1
ss -tln
sudo systemctl --type=service --state=running
sudo find / -perm -4000 -type f 2>/dev/null
curl -I http://127.0.0.1
nmap -sV 127.0.0.1
```

These commands are evidence-gathering examples; actual findings must be recorded from the authorized lab target rather than invented.

1.4 Basic Security Assessment — evidence & remediation

Expected assessment record

Finding ID	Observation	Evidence	Risk / impact	Recommendation
F-01	Observed service/port	Screenshot/log	Explain exposure	Reduce exposure or configure securely
F-02	Permission/config issue	Screenshot/log	Explain unauthorized access possibility	Apply least privilege
F-03	Observed web behavior	HTTP evidence	Explain potential web risk	Validate/fix and retest

Evidence limitation

The supplied internship material contains Linux terminal evidence for user/group work, but it does not contain a complete controlled-VM security assessment result. Therefore, this section documents the assessment method, safe command checklist and evidence format rather than claiming unverified scan results.

Typical finding workflow

Observe → capture evidence → identify the affected service/component → assess security impact → map to a known weakness where appropriate → recommend a fix → retest in the controlled lab.

References

Official technical/security references used for Topics 1.1, 1.3 and 1.4:

- IETF RFC 9293 — Transmission Control Protocol (TCP).
<https://www.rfc-editor.org/rfc/rfc9293.html>
- IETF RFC 768 — User Datagram Protocol (UDP).
<https://www.rfc-editor.org/info/rfc768/>
- IETF RFC 1034 — Domain Names: Concepts and Facilities.
<https://www.rfc-editor.org/rfc/rfc1034.html>
- IETF RFC 1035 — Domain Names: Implementation and Specification.
<https://www.rfc-editor.org/rfc/rfc1035.html>
- IETF RFC 9110 — HTTP Semantics.
<https://www.rfc-editor.org/rfc/rfc9110.html>
- OWASP Top 10:2025.
<https://top10.owasp.org/2025/>
- OWASP Web Security Testing Guide.
<https://wstg.owasp.org/>
- Nmap Official Guide.
<https://nmap.org/book/>

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